

Time-frequency analysis and spiking neural network for motor imagery EEG classification

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Abstract—Accurate recognition of electroencephalogram (EEG) is an important issue in motor imagery (MI)-based brain-computer interface (BCI) system. However, due to the non-linear and low signal-to-noise characteristics of EEG signal, the recognition of MI EEG signal remains a tricky problem. In this paper, a novel framework based on time-frequency analysis and spiking neural network (SNN) is proposed for feature extraction and classification of MI EEG signal. Specifically, the multi-wavelet basis functions (MWBF) are first employed for the approximation of coefficients in the time-varying autoregressive (TVAR) model. An effective regularized orthogonal least squares (ROLS) algorithm is then applied to significantly simplify the model structure. In addition, a power spectral density (PSD) function is defined to obtain high-resolution time-frequency features. Especially, to adequately exploit the spatio-temporal properties of the obtained features, a powerful SNN model is constructed for accurate classification. The proposed method is validated on a widely used dataset, and the results indicate that the classification performance of the proposed method is significantly better than other state-of-the-art methods.

Keywords—motor imagery electroencephalography classification, time-frequency analysis, power spectral density, spiking neural network

I. INTRODUCTION

Brain-computer interface (BCI) is a novel human-machine interface that establishes a direct connection between the brain of human or animal and an external device [1]. BCI technology enables patients with motor dysfunction to control external equipment such as wheelchairs by converting electroencephalography (EEG) signal into corresponding control commands instead of using traditional interfaces such as mice or keyboards. Motor imagery (MI) EEG signal is non-invasive and has the properties of low cost and high temporal resolution. And the MI ability of patients with motor dysfunction is not significantly different from those of normal people, the MI-based BCI system has therefore attracted a great deal of attention and researches in the field of medical rehabilitation. The core problem of BCI system is how to accurately identify the MI EEG signal. Therefore, extracting effective features to achieve high-accuracy classification is the major challenge to be addressed. Currently, the feature extraction methods of MI EEG signal mainly include common space pattern (CSP), time-frequency analysis and deep learning.

CSP is a classical filtering feature extraction method that detects a task-specific signal component from a multi-channel EEG signal. Zang et al. [2] combined wavelet packet decomposition and CSP algorithm to precisely extract the

frequency domain and spatial domain features of EEG signal, and finally realized the valid classification of EEG signals. Meanwhile, CSP algorithm has been subjected to relevant improvements such as the proposed FBCSP and FBRCSP algorithms. However, the drawback of the CSP algorithm is that it is susceptible to the effects of bandwidth frequency and noise.

The commonly available time-frequency analysis methods can be mainly divided into three categories: short-time Fourier transform (STFT), wavelet transform (WT) and parametric spectral estimation methods. Han et al. [3] proposed RCSP-STFT algorithm by effectively integrating STFT and regularized common spatial pattern (RCSP). The EEG signal was first projected into the low-dimensional space by RCSP. STFT was then utilized to obtain the features of fixed frequency bands. However, STFT suffers from the trade-off in time-frequency resolution and spectrum leakage. Abdalsalam et al. [4] derived energy, entropy and power values from the sub-band signals obtained by DWT as features of the EEG signal. However, DWT has a problem in the selection of wavelet basis functions. Li et al. [5] employed a time-varying autoregressive (TVAR) model to model the EEG signal and obtained precise power spectral density (PSD) estimation than the conventional spectral estimation methods by the defined PSD function.

Recently, deep learning method has gained the attention of a large variety of researchers, which can learn higher-level features of MI EEG signal through hierarchical nonlinear mapping. Yang et al. [6] transformed MI EEG signal into 2D images based on the 3D position of electrodes during EEG acquisition, and then proposed an image-based convolutional neural network (CNN) to conduct the classification task of MI EEG signal. However, CNN is less interpretable and ignores the spatio-temporal dynamic information of MI EEG signals. Spiking neural network (SNN) is known as the third generation of artificial neural network (ANN), which is capable of capturing rich spatio-temporal properties. Virgilio et al. [7] employed SNN model to accurately identify MI EEG signal and compared it with other traditional classifiers, and the final results showed that SNN model can achieve better performance with the application of a small number of spiking neurons.

The central aim of this paper is to propose a novel MWBF-ROLS-SNN framework to implement precise recognition of MI EEG signal. Specifically, the multi-wavelet basis functions (MWBF) are first exploited to approximate the time-varying coefficients of the TVAR model. A robust regularized orthogonal least squares (ROLS) algorithm is then applied to significantly refine the structure of the TVAR model and solve

for the associated coefficients. High-resolution time-frequency features can be further derived based on a pre-defined PSD function. To take full advantage of the spatio-temporal dynamic information of the extracted features, an efficient SNN model is employed to accurately classify the MI EEG signals.

The remaining parts of this paper is organized as follows. The identification of the TVAR model, the B-spline wavelet basis functions and SNN model are introduced in detail in section II. The used dataset and associated experiments are presented in section III. The results and discussion are then described in section IV. The conclusion is finally given in section V.

II. METHODOLOGY

A. Identification of the TVAR model based on MWBF and ROLS algorithm

This section focuses on the modeling process of MI EEG signal, which mainly includes the application of the MWBF to model the coefficients of the TVAR model and the employment of the ROLS algorithm to generate a model with sparse structure.

1) *TVAR Model Based on MWBF*: TVAR model is particularly adapted to analyze non-stationary signal such as EEG signal. And a TVAR model of order p is formulated as follows:

$$y(t) = \sum_{i=1}^p \eta_i(t) y(t-i) + \varepsilon(t) \quad (1)$$

where $y(t)$ is the modeled signal, t is the sampled time-series, p is the model order, $\eta_i(t)$ denotes the parameter of the model, $\varepsilon(t)$ represents the Gaussian white noise sequence whose mean is zero and variance is σ_ε^2 . In order to accurately solve the parameter $\eta_i(t)$ of the TVAR model, it can be extended to a series of linear combinations of the basis functions $\varphi_m(t)$, where m is the amount of basis functions. The representation of $\eta_i(t)$ is as follows:

$$\eta_i(t) = \sum_{m=1}^M u_{i,m} \varphi_m(t) \quad (2)$$

where $u_{i,m}$ are the time-invariant coefficients of $\varphi_m(t)$. By defining a new variable $y_m(t-i) = \varphi_m(t) y(t-i)$, the model can then be transformed into a time-invariant linear AR model with the following expression.

$$y(t) = \sum_{i=1}^p \sum_{m=1}^M u_{i,m} y_m(t-i) + \varepsilon(t) \quad (3)$$

Defining Y and Γ to denote the corresponding regression terms and coefficient vectors separately, the matrix form of (3) can be indicated as follows:

$$y = Y\Gamma + \varepsilon \quad (4)$$

2) *B-spline Wavelet Basis Functions*: B-spline basis functions have the advantages of compact support, multi-resolution analysis and apparent analytic form, which make B-splines more applicable to the modeling of non-stationary signals relative to other basis functions. The defined value range of the n -th order B-spline is $[0, n]$, and it is assumed that the function to be estimated with decomposition is

defined within $[0, 1]$. Then for any specified scale index j , the shift index k takes values in $T_n = \{k: -n \leq k \leq 2^j - 1\}$. In most cases, the ideal order of B-splines is 2 to 4, and the preferred value of j is 3. Thus the parameters $\eta_i(t)$ in (1) can be composed of a linear combination of B-splines, which is shown as follows:

$$\eta_i(t) = \sum_{n=2}^4 \sum_{k \in T_n} u_{i,k}^{(n)} \psi_{j,k}^{(n)}\left(\frac{t}{N}\right) \quad (5)$$

where $\psi_{j,k}^{(n)}\left(\frac{t}{N}\right) = 2^{j/2} B_n(2^j x - k)$, and if $\{\varphi_m(t)\}$ in (2) is substituted by $\{\psi_{j,k}^{(n)}\left(\frac{t}{N}\right)\}$, then (5) can be easily converted into (2).

3) *Model simplification and parameter estimation*: Recently, for dynamic regression problems that contain a huge amount of candidate terms, the ROLS algorithm based on the error reduction rate (ERR) criterion is proven to be effective for model simplification and parameter estimation. The overall workflow of the ROLS algorithm is as follows: Step1: Construct a suitable linear TVAR model in (1) with real data.

Step2: Approximate each time-varying coefficient employing B-splines.

Step3: Solve the regularized error reduction ratio (RERR) values of all candidate terms, and choose the term that has the maximum RERR value as the initial term and then eliminate it from the candidate terms.

Step4: Proceed to q -th step, orthogonalize each remaining term with the previous $q-1$ terms, and pick the one with the maximum RERR value as the q -th term.

Step5: Count the grand total of all RERR values. Abort the forward regression process if the value of $1 - \sum_{u=1}^q RERR_u$ is less than the set threshold; otherwise, repeat step4 until the condition is satisfied.

Step6: Estimate the associated parameters of the selected model term and solve for the initial time-varying coefficients in (5).

B. High-Resolution time-frequency PSD estimation

Once the TVAR model is effectively simplified and the model coefficients are obtained, the high-resolution PSD estimation is defined as follows:

$$\text{PSD}(t, f) = \frac{\hat{\sigma}_\varepsilon^2}{\left| 1 - \sum_{i=1}^p \hat{\eta}_i(t) e^{-j2\pi i f / f_s} \right|^2} \quad (6)$$

where f_s means the sampling frequency, $\hat{\eta}_i(t)$ and $\hat{\sigma}_\varepsilon^2$ denote the estimated values of $\eta_i(t)$ and σ_ε^2 separately, j takes the value of $\sqrt{-1}$.

The effective time-frequency features of the EEG signal are able to derived from the obtained PSD estimation. And the features of specific time and frequency bands are defined as follows:

$$F(b, d) = \int_{t_b} \int_{f_d} \text{PSD}_Y(t, f) df dt \quad (7)$$

where PSD_Y is PSD of the estimated signal Y .

C. SNN model

Previous research mainly focused on using random forest (RF), support vector machine (SVM) and ANN for classification tasks. However, these classifiers are more applicable to deal with static features. SNN not only retains the promising characteristics of traditional ANN, but also achieves a higher level of biological neuron simulation, which is able to capture rich spatio-temporal dynamic features relative to traditional classifiers. This section mainly describes two neuron models, Leaky Integrate and Fire (LIF) and Leaky Integrator (LI), and the basic structure of SNN model, each of which is described in detail as follows.

1) *LIF and LI neuron models* : Spiking neuron is the basic unit of SNN model and determine the underlying dynamical properties of SNN model. Different types of spiking neurons describe the dynamic changes of potentials during neuronal work at different levels of detail. Compared with other spiking neurons, the mathematical models of LIF and LI neurons are simpler, more computationally efficient and able to capture the dynamic properties of the network inputs. Therefore, LIF and LI neurons are widely used as the basic units of SNN model. LIF neuron can be considered as an equivalent RC circuit, which is a combination of a resistor and a capacitor in parallel. According to the basic principle of the circuit, the 1-order differential equation of the membrane potential change is as follows:

$$I(t) = \frac{u(t) - u_{reset}}{R} + C \frac{du}{dt} \quad (8)$$

Fig. 1 presents the structural diagrams of traditional McCulloch-Pitts (MP) and LIF neuron models. The most essential difference of the two neurons is the mechanism of information processing. It is obvious that the MP neuron takes real values as inputs and outputs, which greatly simplifies the biological neuron. The LIF neuron, however, takes spike sequences carrying temporal information as inputs and outputs, which makes it closer to the real biological neuron. Meanwhile, the LI neuron works in the same way as LIF neuron except that it does not fire spike sequences.

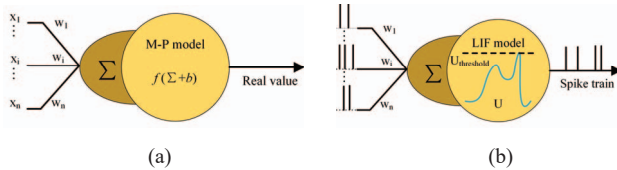


Fig. 1. Structural diagrams of different neuron models. (a) M-P neuron. (b) LIF neuron.

2) *The basic structure of SNN model*: Fig. 2 presents the structure of SNN model. Specifically, SNN model first takes binary spike sequences as the inputs. It is notable that if the inputs of the network are non-spike stream data, an additional encoding layer should be added to encode the input data. Then one or more layers of LIF neurons are defined in the hidden layer, which are utilized to powerfully extract the spatio-temporal feature information of the input spike sequences. Finally, the output layer records the changes of neuronal membrane potential and the number of spike fired in all time steps, based on which the most active neuron can be found and the final output results can be determined.

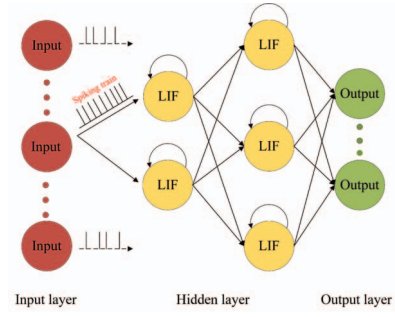


Fig. 2. Schematic diagram of SNN model structure.

III. DATASET AND EXPERIMENTS

A. Dataset description

The dataset utilized in this paper is dataset III, which was provided by Graz University of Technology in BCI competition II. Fig. 3(a) illustrates the experimental paradigm. The subject was in a quiet and relaxed state for the first 2 seconds. At the 2nd second, an audible signal appeared to indicate the start of the experiment. At the same time a "+" was presented on the screen and remained there for 1 second. At the 3rd second, an arrow appeared and the subject imagined the movement of the hands according to the point of the arrow until the end of the 9th second. The acquisition channels of the signals were C3, Cz and C4, and Fig. 3(b) shows the electrode positions. The entire experiment was composed of 280 groups, 140 groups each for left-handed and right-handed tasks. And the signal was sample-rated at 128 Hz with band-pass filtering between 0.5 and 30 Hz.

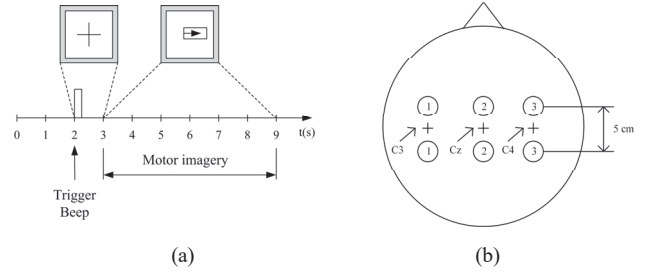


Fig. 3. Schematic diagrams of experimental paradigm and electrode positions. (a) Experimental paradigm. (b) Electrode positions.

B. Experiments

1) *Time-frequency features of the inputs to SNN model*: The activity in motor-related areas of the brain is mainly reflected in Mu (8-13 Hz) and Beta (18-26 Hz) rhythms. These two frequency bands are therefore used for the next analysis. And the frequency band of Beta rhythm is extended to 14-26 Hz in order to prevent the loss of effective features. In this experiment, the 3-8 second signal is selected and divided into 5 equal segments for feature extraction. It can be seen that a signal is partitioned into 10 time-frequency regions and thus 10 features can be obtained. Meanwhile, since Cz channel contains little MI-based information, only the signals of channels C3 and C4 are employed for the classification task. Therefore, there are a total of 20 dimensional features for each sample group.

2) *The structure of the proposed SNN model*: SNN model is built and trained based on Norse and PyTorch in this experiment. Fig. 4 displays the overall structure of the proposed SNN model which is utilized for MI EEG

classification. Firstly, the proposed SNN model takes 20 dimensional non-spike time-frequency features as inputs. Since SNN is only capable of information processing for spike sequences, an encoding layer based on M-P and LIF neurons is constructed to encode the non-spike data. The generated spike sequences are then further transmitted to the LI neuron-based readout layer, which can record the membrane potential changes of neurons representing different classes. Finally, the decoding layer is designed based on the softmax function. MI EEG signal is efficiently recognised by calculating the maximum value of the membrane potential in the time dimension. The time step parameter of the proposed SNN model is 48, and SNN model is trained with batchsize of 20 and epoch of 100.

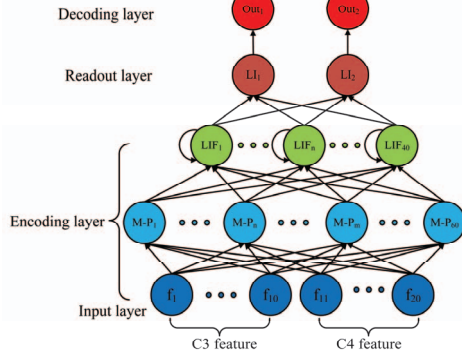


Fig. 4. The structure of the proposed SNN model for MI EEG classification.

3) *Performance evaluation*: To ascertain the reliability of the model, all experiments are subjected to 10-fold cross-validation. Meanwhile, Accuracy and Kappa metrics are applied to evaluate the classification performance of the model. These two metrics are defined separately as follows.

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN} \quad (9)$$

$$Kappa = \frac{Accuracy - p_e}{1 - p_e} \quad (10)$$

where TP and TN, FP and FN, represent the samples correctly and incorrectly classified by the model, respectively. And $p_e = \frac{a_1 \times b_1 + a_2 \times b_2 + \dots + a_c \times b_c}{n \times n}$, where n is the total amount of samples, $a_1, a_2, \dots, a_c$ and $b_1, b_2, \dots, b_c$ stand for the amount of true samples and predicted samples in each class, respectively.

IV. RESULTS AND DISCUSSION

In this section, the PSD energy distribution based on different methods is first presented. The classification effects of the proposed SNN model and some traditional machine learning classifiers are then shown. A comparison is finally conducted between the proposed MWBF-ROLS-SNN method and the latest excellent methods.

A. Time-frequency PSD distribution based on different methods

In this experiment, a 4-order TVAR model is utilized to approximate the MI EEG signal, and once a sparse TVAR model is determined based on the ROLS algorithm, high-resolution time-frequency energy distribution is capable of being obtained according to (6). Fig. 5 presents the PSD distribution derived with the proposed MWBF-ROLS-SNN method, and non-parametric spectral estimation methods such as STFT and CWT are used for comparison. It is obvious from Fig. 5 that the STFT and CWT methods are not able to simultaneously realize high resolution in the time and frequency domains. However, the proposed MWBF-ROLS-SNN method greatly improves the time-frequency resolution and can lead to a more detailed energy distribution. This is mainly due to the proposed MWBF-ROLS-SNN method that introduces the MWBF and ROLS algorithm, which enable the model capture the instantaneous information of MI EEG signal and thus yield a higher time-frequency resolution. Meanwhile, it can be clearly seen that the energy difference between C3 and C4 channels in Mu rhythm is highly evident in the energy distribution obtained by the proposed MWBF-ROLS-SNN method, which is called event-related synchronization (ERS) or event-related desynchronization (ERD) phenomenon.

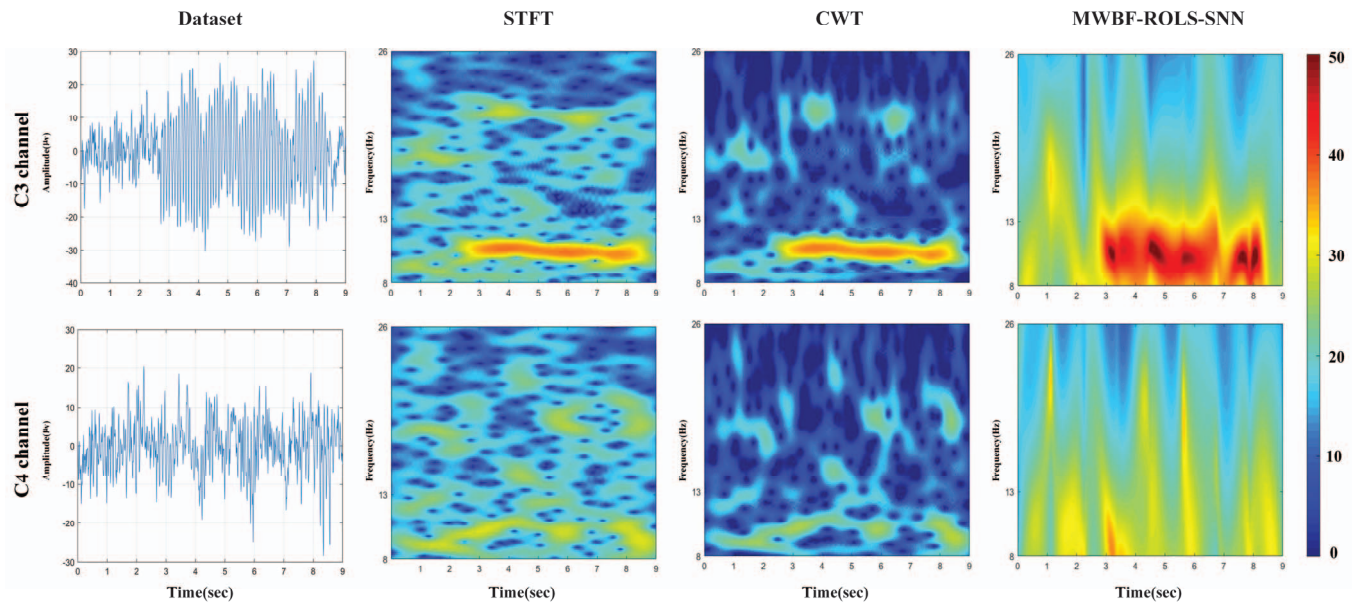


Fig. 5. Time-frequency PSD distribution obtained using different methods.

B. Classification effects of the Proposed SNN Model

To highlight the superior properties of the constructed SNN model, Table I presents the classification results of SNN model and some traditional machine learning classifiers based on the same time-frequency features. It is obvious from Table I that the constructed SNN model produces optimal results relative to other classifiers in both accuracy and kappa metrics. Specifically, the RF and XGBoost classifiers produce almost identical recognition results with accuracy and kappa values of 87.5% and 0.750, respectively. The SVM classifier achieves an efficient improvement relative to the RF and XGBoost classifiers, reaching an accuracy of 91.8% and a kappa value of 0.836. The accuracy and kappa values of ANN are 90.4% and 0.808, which are in between the classification results of SVM and RF classifiers. Notably, the proposed SNN model improves the accuracy and kappa values to 93.9% and 0.878, which are significant improvements compared to other classifiers. Therefore, it is demonstrated that the SNN model can sufficiently utilize the spatio-temporal characteristics of the input data and has excellent classification performance.

TABLE I. THE PERFORMANCE OF DIFFERENT CLASSIFIERS

Classifier	Evaluation Metric	
	Accuracy(%)	Kappa
RF	87.5	0.750
XGBoost	87.5	0.750
SVM	91.8	0.836
ANN	90.4	0.808
SNN	93.9	0.878

C. Performance comparison with Previous Studies

To demonstrate the validity of the proposed MWBF-ROLS-SNN method, Table II illustrates the performance comparison between the proposed method and other studies for the same dataset. As shown in Table II, the proposed method obtains promising result relative to other studies. The superior characteristics of the proposed method can be attributed to four aspects: 1) MWBF are adopted to solve the time-varying coefficients of the TVAR model; 2) the ROLS algorithm is employed to determine a streamlined model; 3) the PSD function is defined to generate high-resolution time-frequency features; 4) a dynamic SNN model is applied to distinguish the input features efficiently. Therefore, the proposed MWBF-ROLS-SNN method is competent for accurate identification of MI EEG signal.

TABLE II. PERFORMANCE COMPARISON OF THE PROPOSED METHOD WITH EXISTING METHODS

Proposed By	Method	Classifier	Accuracy (%)
Kim et al. [8]	CWT	CNN	90.0
Eslahi et al. [9]	Genetic Algorithm	LDA	84.0
Tabar et al. [10]	STFT	CNN-SAE	90.0
Li et al. [11]	DWT and average power	BP	92.4
Proposed method	MWBF-ROLS-SNN	SNN	93.9

V. CONCLUSION

In this paper, a new MWBF-ROLS-SNN framework is proposed for the modeling and identification of MI EEG signal. There are mainly four contributions in this paper: 1) MWBF are introduced for parametric modeling in the process of constructing TVAR model; 2) the outstanding ROLS algorithm is employed to effectively optimize the model; 3) the PSD function is applied to generate highly discriminative time-frequency features; 4) SNN, which can highly exploit the spatio-temporal characteristics of the input features, is utilized to solve the classification problem. The promising results in this paper indicate that the proposed MWBF-ROLS-SNN method is accurate, efficient and interpretable. In addition, the neurons of SNN model are activated only when spikes are fired. Based on this characteristic, SNN model is able to perform complex data processing with ultra-low power consumption, which contributes significantly to the construction of low-latency, low-power and high-accuracy MI-based BCI system. Therefore, the proposed MWBF-ROLS-SNN method is expected to be better applied in the future.

ACKNOWLEDGMENT

We are grateful to the Graz University of Technology for providing the public dataset.

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